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A systematic review of microwave-based therapy for axillary hyperhidrosis

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ABSTRACT

Objective: To systematically analyse the literature on the use of the microwave-based device for subdermal thermolysis of the axilla and its efficacy for the treatment of axillary hyperhidrosis.

Materials and methods: A systematic review was conducted using PubMed, Embase, SCOPUS and Cochrane databases on 2 June 2016. The inclusion criteria including: (1) studies with human subjects, (2) full-text articles published in English, (3) a microwave-based device used to treat axillary hyperhidrosis and (4) trials that precisely evaluated axillary hyperhidrosis. Exclusion criteria were the following: (1) studies that did not fit the inclusion criteria mentioned above and (2) case reports and reviews.

Results: We reviewed five clinical trials and 189 patients, all of which were published between 2012 and 2016. There was one randomized controlled trial, one retrospective study and the remainder were prospective studies. Although all of the studies were conducted with a small sample size, the results indicated that microwave-based device treatment of axillary hyperhidrosis had long-term efficacy with mild adverse effects. In addition, most patients were satisfied with the outcomes in these studies.

Conclusions: Microwave-based device treatment may be an effective alternative treatment for axillary hyperhidrosis. However, further investigation is necessary to determine its long-term efficacy and safety.

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KEYWORDS
Axillary hyperhidrosis; bromhidrosis; hyperhidrosis; microwave; osmidrosis

Introduction

Axillary hyperhidrosis is a common problem presenting in 73% of patients with hyperhidrosis and is found in 1.4% of the population of the USA (1,2). The precise mechanism of the condition is not fully understood, and overstimulation of the eccrine sweat glands is thought to be the cause. Axillary hyperhidrosis is not a life-threatening disease, but it can be a substantial burden, affecting relationships with other people, leisure activities, emotional well-being and self-esteem (3). There are several treatments of axillary hyperhidrosis, including topical and injectable agents, oral medication, surgical methods and laser treatment (4). However, non-invasive treatments only have temporary effects, and invasive treatments are limited by their complications (5). Although good efficacy has been reported for laser treatment, insertion of the laser fibre into the subdermal space may lead to complications if the procedure is not well conducted (6). The microwave-based device, which was introduced by Johnson and colleagues in 2012, has been approved by the United States Food and Drug Administration (7). Using a microwaves-based device, the water-rich dermis and sweat glands can be selectively heated by the physical rotation of high dipole molecules. A prior randomized, blinded, multicentre study also proved the high efficacy of microwave technology (8). However, other studies were reported using a small sample size, and there are no published systematic reviews that have evaluated this method for the treatment of axillary hyperhidrosis. The aim of this study was to systematically synthesize the literature on the use of a microwave-based device and its efficacy for the treatment of axillary hyperhidrosis.

Methods

Data sources

The PubMed, Embase, Scopus and Cochrane databases were searched for studies published during the period up to 2 June 2016. The following medical search headings terms, words and combinations of words were used in the systematic search: 'bromhidrosis' or 'osmidrosis' or 'hyperhidrosis' and 'axilla' or 'axillae' or 'axillary'. All included studies were also entered into the PubMed 'similar articles' function and the science citation index. Moreover, we identified additional studies by manually searching the reference sections of these papers and by contacting known experts in the field. Finally, unpublished trials were retrieved from the ClinicalTrials.gov registry (http://clinicaltrials.gov/). This systematic review has been accepted by Prospero, an online international prospective register of systematic reviews that is curated by the National Institute for Health Research (CRD42016039886).

Study selection and data extraction

The following studies were included in the analysis: (1) studies with human subjects; (2) full-text articles published in English; (3) microwave-based device used to treat axillary hyperhidrosis; and (4) trials that precisely evaluated axillary

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hyperhidrosis. Exclusion criteria were: (1) studies that did not fit the inclusion criteria mentioned above and (2) case reports and reviews.

Two independent reviewers (T. H. Hsu and Y. K. Tu) extracted the data from the trials, including the participants, inclusion and exclusion criteria, administration of therapy, prevalence and assessment of axillary hyperhidrosis, and complications. Discrepancies and any disagreements were resolved through discussion with a third reviewer (C. N. Lee). The authors of the studies were contacted for additional information when necessary.

Data synthesis and analysis

The identified studies were reviewed systematically. The following information was extracted from each study: year and regions, study design, interventions, follow-up duration, treatment protocol, treatment contents, major inclusion/exclusion criteria, treatment, baseline patient characteristics, subjective and objective outcomes measures, results and adverse events.

Results

Characteristics of the trials

The review process is outlined in Figure 1. The initial search yielded 2012 studies, 1985 of which were deemed ineligible through screening of titles and abstracts. Subsequently, the full text of 37 studies was screened. Of these, ten were full text not published, nine were excluded because they were review articles, six did not meet the eligibility criteria because of duplicate publication, four did not use microwave treatment, and three did not treat axillary hyperhidrosis. Thus, five eligible trials were included in this systematic review (8–12).

The main characteristics of the analysed studies are listed in Table 1, all of which were published between 2012 and 2016, except for one study that presented results at different follow-up times (12). The studies were conducted in different regions, including Australia, Canada, Korea, Taiwan and the United States. Three of the five studies were prospective, observational studies that included only microwave-based device treatment, and one was a retrospective study and the rest one was randomized controlled trial (RCT) that included both microwave-based devices and placebo. Most patients accepted more than one treatment; however, in one study (10), all subjects underwent one microwave treatment for each axilla. The sample size for the microwave device arm varied substantially, ranging from 7 to 81 patients (14–162 axillae). However, the majority (four studies) had a sample size of no more than 31 patients in the microwave groups. Three studies had a 12-month, one study had 7-month and the rest had a 3-month follow-up period. All the studies used the microwave-based device; two studies used miraDry System; one used DTS G2 System; the remaining two did not describe a system for a microwave device, but they described a frequency between 300 MHz and 300 GHz in the treatment.

All the studies included only adults and required patients to have symptomatic axillae of either hyperhidrosis or osmidrosis (Table 2). The median age of participants in the studies was between 30 and 37 years, although the age range varied among each study. Patients with axillary hyperhidrosis were all evaluated by the Hyperhidrosis Disease Severity Scale (HDSS) and had an HDSS score of 3 or 4. Glaser et al. and Hong et al. used the gravimetric reading of axillary hyperhidrosis, which is an objective measure, and the patients had a mean baseline gravimetric reading of 179.4 and 190 mg/5 min, respectively. Three studies excluded patients who treated their axillary hyperhidrosis with surgery in the past or with botulinum toxin injection 12 months before the study. On the other hand, one study included patients in which previous

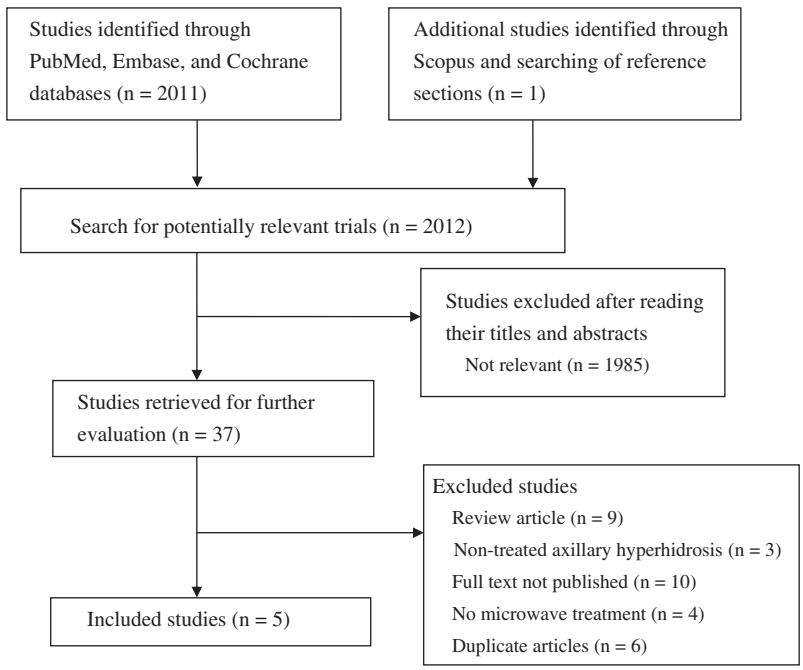

Figure 1. Flowchart for study selection.

Table 1. Study articles.

Author [Year]	Region	Study design	Intervention	Follow-up duration	Treatment protocol	Treatment content
Glaser et al (8) [2012]	United States	Randomized controlled placebo-controlled trial	Microwave group: $n = 81$ (162 axillae) control group: $n = 39$ (78 axillae)	Microwave group: 3, 6, 9 and 12 mo Control group: 3 and 6 mo	Three procedure sessions were administered at about 2-wk interval. Fewer procedure sessions were allowed if the subject's underarm sweat was eliminated with each session or if the subject declined.	Microwave group: Marking the axilla starch–iodine test (An alcohol-based iodine mixture was wiped on the skin of the axilla, and then corn starch was sprinkled on the area) Local anaesthesia 1% lidocaine with 1:100,000 epinephrine Microwave device the device included integrated vacuum and cooling (DTS G2 System; Miramar Labs, Sunnyvale, CA). Control group: all steps were same as above, but microwave energy was not delivered.
Hong et al (9) [2012]	Canada	Prospective single-arm observational study	Microwave group: $n = 31$	1, 3, 6 and 12 mo after their last procedure session.	The treatment started with the first procedure visit. Subsequent treatments were held approximately every 30 d after sweat assessments and safety evaluation. At most, three procedure sessions were allowed, and all procedures completed within a 6-month window.	Marking the axilla starch–iodine test (An alcohol-based iodine mixture was wiped on the skin of the axilla, and then corn starch was sprinkled on the area) Local anaesthesia 1% lidocaine with 1:100,000 epinephrine Microwave device the device included integrated vacuum and cooling (miraDry System; Miramar Labs, Sunnyvale, CA) and allowed for ±10% of energy settings.
Lee et al (10) [2013]	Korea	Prospective single-arm observational study	Microwave group: $n = 11$	7 mo after treatment	All subjects underwent one microwave treatment on each axilla.	Marking the axilla starch–iodine test Local anaesthesia mixture of normal saline, 100 cc; epinephrine, 0.2 cc; 2% lidocaine, 10 cc; and bivon, 1 cc Microwave device an electromagnetic signal with a frequency between 300 MHz and 300 GHz
Chang et al (11) [2014]	Taiwan	Prospective single-arm observational study	Microwave group: $n = 7$	1 and 3 mo after treatment	Two procedure sessions were administered at 3 mo intervals.	Marking the axilla not known Local anaesthesia 1% lidocaine with 1:100,000 epinephrine Microwave device the device included integrated vacuum and cooling (miraDry system; Miramar Labs, Santa Clara, CA, USA); Energy levels 3 and 5 were set for first and second treatment
Scuderi et al (12) [2016]	Australia	Retrospective single-arm observational study	Microwave group: $n = 20$	Average 5 mo (range from 1 to 12 mo)	All subjects underwent at least one microwave treatment. Fifteen patients returned for the second treatment at around 3 mo	Marking the axilla starch–iodine test Local anaesthesia 0.05% lignocaine Microwave device The machine emits a constant frequency of 5800 MHz Most of the patients were treated with Level 3 or 4 at the initial visit and Level 4 or 5 at the final visit (Level from 1 to 5 increased with longer period and more heat)

d, days; mo, months; SD, standard deviation.

Table 2. Characteristics of the included studies.

Study [Year]	Inclusion criteria/exclusion criteria	Patient baseline characteristics
Glaser et al (8) [2012]	(1) Inclusion criteria: Subjects had a HDSS score of 3 or 4 and had baseline axillary sweat production of greater than 50 mg/5 min as measured by gravimetric readings (2) Exclusion criteria: Subjects had prior surgery or botulinum toxin injections to treat PAH within the past 12 mo	Microwave group No. of patients (male %): 81 (33) Median age: 33 y HDSS score (No. of patients): 3 (41), 4 (40) Mean baseline gravimetric reading: 180.5 mg/5 min Control group No. of patients (male %): 39 (47) Median age: 31 y HDSS score: 3 = 26, 4 = 13 Mean baseline gravimetric reading: 177.2 mg/5 min
Hong et al (9) [2012]	(1) Inclusion criteria: Subjects had a HDSS score of 3 or 4 and had baseline axillary sweat production of greater than 50 mg/5 min as measured by gravimetric readings (2) Exclusion criteria: Subjects had prior surgery or botulinum toxin injections to treat PAH within the past 12 mo	No. of patients (male %): 31 (26) Median age (range): 33 y (18–65) HDSS score (No. of patients): 3 (20), 4 (11) Mean baseline gravimetric reading: 190 mg/5 min Mean baseline dermatologic life quality index: 11.8
Lee et al (10) [2013]	(1) Inclusion criteria: Patients with significant axillary hyperhidrosis or osmidrosis were recruited (2) Exclusion criteria: Not described	No. of patients (male %): 11 (27) Median age (range): 37 y (20–52) Diagnosis: Hyperhidrosis = 3, Osmidrosis = 5, both = 3 HDSS score (No. of patients): 3 (3), 4 (3) Odor grade (No. of patients): 3 (1), 4 (7)
Chang et al (11) [2014]	(1) Inclusion criteria: Subjects had a HDSS score of 3 or 4 and at least 5 on the Odor-10 scale. (2) Exclusion criteria: No patient had been treated with endoscopic thoracic sympathectomy, liposuction or other surgeries for axillary osmidrosis or hyperhidrosis. They had no botulinum toxin injection within 1 year and had not used any topical treatment within 14 d prior to treatment.	No. of patients (male %): 7 (2.8) Median age (range): 31 y (22–53) Mean BMI (kg/m^2): 21.1 HDSS score (No. of patients): 3 (2), 4 (5) Odor-10 grade (No. of patients): 5 (1), 6 (1), 8 (2), 9 (2), 10 (1)
Scuderi et al (12) [2016]	(1) Inclusion criteria: patients with HDSS ratings three and above (2) Exclusion criteria: not described	No. of patients (male %): 20 (55) Median age (range): 30 y (19–45) Mean HDSS score: 3.75 ± 0.44 All patients had failed previous treatments with topical agents.

d, days; mo, months; y, years old; DLQI, Dermatologic Life Quality Index; HDSS, Hyperhidrosis Disease Severity Scale; PAH, primary axillary hyperhidrosis.

treatments with topical agents had failed, and the remaining one did not mention the previous treatment of patients.

The administration of a microwave-based device also varied regarding frequency and treatment intervals. The intervention ranged from a single administration to three times during the course of treatment. The interval ranged from two weeks to three months if multiple administrations were implemented.

With respect to the outcomes assessed in these studies, all studies included adverse events and function measures (Table 3). All studies included HDSS, two also used a gravimetric assessment for axillary hyperhidrosis, and one study measured sweat reduction by the patient's estimates. In addition, three studies also evaluated osmidrosis with subject-reported odour evaluation, odour grade and odour-10 score, respectively. Only one study used Dermatologic Life Quality Index (DLQI) to evaluate quality of life. Histological studies were also performed in three studies, and only one study had a full description of the results of these histological analyses (11). Two studies obtained patient satisfaction, and only one study used a pain scale to evaluate postoperative pain.

Outcome associated with microwave-based device therapy

Regardless of the outcome measures, all studies demonstrated the efficacy of microwave therapy for the reduction of sweating among patients with axillary hyperhidrosis (Table 3). The onset of the effects of treatment was not described in full detail in all studies (efficacy prior to the completion of the procedure was not described); however, it appears that the effects exhibited long-term stability. Subjective outcome HDSS in all studies demonstrated at least a one-point decrease in most patients throughout the follow-up duration. In addition, there was at least a two-point drop in HDSS that was also revealed in some studies (9–11); 86% (11), 83.3% (10) and 55% (9) of subjects had at least a two-point decrease at the 3-, 7- and 12-month follow-up visit, respectively. There were two studies that conducted statistical analyses in HDSS. In the only RCT, the percentage of subjects that decreased by at least two points in the HDSS in microwave group was statistically greater than that in the control group through the six-month follow-up period (8). The other study found that the post-treatment mean HDSS points were statistically significantly lower than the points for the pretreatment (12). The objective outcome gravimetric assessment was performed in two studies. In one study, the authors found that there was at least 80 and 62% patients who had no less than 50 and 75% sweat reduction, respectively, at the 30-day visits (8). The results reached statistical significance between the microwave and control group when comparing the proportion of patients who had at least a 75% sweat reduction (8). However, there was no statistical significance at the later time points up to six months (8). In another study, at least 90% of patients had greater that 50% sweat reduction, and the average sweat reduction was at least 82% during the 12-month follow-up visits (9). Moreover, in one study, patients estimated sweat

Table 3. Outcome measures and results.

Study[Year]	Subjective outcome	Objective outcome	Results	Adverse effects
Glaser et al (8) [2012]	HDSS	Gravimetric assessment	HDSS: At 30 d, HDSS in 89% of microwave group and 54% of control group were decreased to 1 or 2 ($p < 0.001$). Through the 6 months, the HDSS efficacy (HDSS decreased ≥2 or HDSS score = 1 or 2) for the microwave group was statistically significantly greater than the control group. One year after treatment, the HDSS efficacy of microwave group continued to be stable. Gravimetric assessment: At 30 d, a reduction ≧ 50% in sweat was seen in 80% of patients in microwave group and 67% of control group ($p = 0.07$); a reduction ≧ 75% sweat was seen in 62% of patients in microwave group and 39% in control group ($p = 0.01$). At later time points, there was no statistically significant difference in gravimetrically measured sweat reduction between the two groups	There were no procedure-related serious adverse events reported in the study for any subject. Altered sensation, pain/soreness, swelling, blister/burn/ulceration, skin rash/irritation/dermatitis, axillary bumps/nodules and compensatory sweating were reported in these studies. The most frequently reported adverse event (9.9% of treated subjects) was altered sensation in a moderately sized area (average 12-cm length at onset). All the adverse effects disappeared within 30 d in average except for altered sensation (with mean 67.4 d).
Hong et al (9) [2012]	HDSS DLQI Satisfaction questionnaire Subject-reported odour evaluation	Gravimetric assessment Biopsy	HDSS: At all follow-up visits, more than 90% of patients had decreased HDSS score ≧ 1point. At the final 12-mo visit, 94% (29/31) of the subjects had at least a 1-point drop in HDSS and 55% (17/31) had a 2-point drop or greater in HDSS. Gravimetric assessment: At all follow-up visits, ≧ 90% of patients had 50% or greater reduction in gravimetric assessment. The average reduction in sweat for all patients was 83, 82, 82 and 82% at the 1-, 3-, 6- and 12-mo visits, respectively. Biopsy: A sample was taken in one axilla 105 d after the last treatment, with no sweat glands evident. DLQI: There were at least 5 points reduction in 96, 89, 89 and 85% of subjects at the 1-, 3-, 6- and 12-mo visits, respectively; the average DLQI reduction was 10.4, 10.2, 9.6 and 9.9 points for the 1-, 3-, 6- and 12-mo visits, respectively. Satisfaction questionnaire: There were 90, 96, 93 and 88.5% subjects reported being very satisfied at 1-, 3-, 6- and 12-mo visits Subject-reported odour evaluation: There were 67.7, 71.0, 74.2 and 61.3% of subjects who had no noticeable odour (baseline was 12.9%) at 1-, 3-, 6- and 12-mo visits	There were no procedure-related serious adverse events reported, and all adverse effects were resolved. The most common were edema, redness and vacuum acquisition mark, and discomfort in 90, 87 and 84% of patients, respectively. The longer-term effects were altered sensation in the skin (median duration was 37 d, range 4 d to 4 mo) and palpable bumps (median duration was 41 d) in 65 and 71% patients, respectively. Axillary hair loss was reported by 26% of subjects. One subject experienced transient neuropathy which resolved at 6 mo.
Lee et al (10) [2013]	HDSS Odour grade (with 4 grades)	Biopsy	HDSS: At the 7-month follow-up visit, there were 83.3% (10/12 axillae) of subjects who had at least a 2-point drop. Odour grade: There were 93.8% (15/16 axillae) of subjects who had at least a 1-grade drop, and 91.3% (13/16 axillae) of subjects who had at least a 2-grade drop. Biopsy: Six months after treatment, specimens from the axillary skin of one patient showed no sweat glands.	One patient complained of loss of sensation and fully recovered in 3 mo. One patient reported skin sagging in both arms that disappeared after 2 mo. One patient had hyperpigmentation, which resolved after 2 mo. Patients experienced transient adverse effects such as pain ($n = 2$), tension ($n = 2$), edema ($n = 3$) and nodules ($n = 1$), which all disappeared within 10 d.

(*Continued*)

Table 3. (Continued).

Study[Year]	Subjective outcome	Objective outcome	Results	Adverse effects
Chang et al (11) [2014]	HDSS Odour-10 score patients satisfaction pain scale	Biopsy	HDSS: At 30 days, 57 and 86% of subjects had at least a 2-point and 1-point drop, respectively, and one person had no drop point after treatment. At 90 days, 86 and 100% of subjects had at least a 2-point and 1-point drop, respectively. Odour-10 score: The mean percentage of reduction in odour-10 scale was 76.4% and 61.8% at 30-day and 90-day visits, respectively. Biopsy: Mean reduction in the area of apocrine sweat glands was 9.61 mm^2 (95% confidence interval, 6.35–12.87 mm^2) and 93% (95% confidence interval, 86.3–99.7%) in percentage. There were significant differences between pre- and postoperative AAG and AAG/TAT ratio ($p = 0.001$ and $p < 0.001$, respectively), which represented a substantial decrease of apocrine glands. Patients satisfaction: Four out of seven participants were very satisfied, two participants were satisfied, and only one patient was unsatisfied.	Pain scale: Postoperative median pain scales were 7 and 6 in first and second procedure session, respectively, and both decreased no more than 2 points in postoperative day 6. Other common adverse events included local erythema, bruise, local swelling, numbness and hypotrichosis. Erythema and bruises were presented in all patients and resolved in a few days. Local swelling mostly went away in 1–2 weeks. Three patients reported hypotrichosis and decelerated axillary hair growth rate, which existed up to their 90-d follow-up visit. No severe and debilitating adverse effects were observed. All patients showed good tolerance during all treatment and follow-up visits
Scuderi et al (12) [2016]	HDSS Standard questionnaire		Mean HDSS: pretreatment = 3.75 ± 0.44 posttreatment = 1.75 ± 0.55 ($p < 0.0001$) Standard questionnaire: mean sweat reduction based on the patient's estimation = 72.5% ± 14.82 The average estimated sweat reduction of the short-term (3 mo or less) and long-term (more than 3 mo and up to 12 mo) follow-up group was 74.3 and 77.7%, respectively.	Short-term (% of patients): pain (65%), swelling (60%), bruising (15%) and temporary numbness (20%) Long-term (% of patients): hair loss (25%) and nodule formation (25%) The average return to complete physical activity was 8.75 d after their last treatment

d, days; mo, months; AAG, area of tissue containing apocrine gland; DLQI, Dermatologic Life Quality Index; HDSS, Hyperhidrosis Disease Severity Scale; TAT, total area of the tissue.

reduction based on a questionnaire, and the results revealed an average of 74.3 and 77.7% sweat reduction in the short-term and long-term follow-up visits, respectively (12).

With the exception of hyperhidrosis, the odour was also evaluated in the three studies (9–11). There was 61.3–74.2% of patients who did not notice the odour (baseline was 12.9%) during the 12-month follow-up visits in one study (9). The other study found that 91.3% of subjects had at least a two-grade drop in the odour grade (10). The remaining studies used an odour-10 score to evaluate the odour, and the mean percentage of reduction in the odour-10 scale was 76.4 and 61.8% for the 30-day and 90-day visits, respectively (11).

Histologic examinations were performed in three studies (9–11). Two of the studies took specimens from the axillary skin of one patient separately, and both results revealed the absence of sweat glands (9,10). The remaining study performed a histologic evaluation for every patient, and the results showed average 9.61 mm^2 and 93% of the reduction in the area of the apocrine sweat glands (11). The pre- and postoperative AAG and AAG/TAT ratio, which represented a substantial decrease of apocrine glands, also achieved a statistical difference in the same study (11).

Other outcomes measurements were also performed in the included studies. In one study, DLQI was used to evaluate the quality of life following treatment (9). The mean reduction of DLQI was at least 9.9 points throughout the 12-month visits. More than 85% of patients were satisfied after treatment based on the different satisfaction assessments of the two studies (9,11).

Adverse events

Short-term adverse events, including pain, edema, blister/burn/ulceration, rash/irritation/dermatitis, tension, numbness and compensatory sweating were reported in the analysed studies, and most events resolved within 30 days. Long-term adverse events, including altered sensation, skin sagging, hypotrichosis, hair loss, hyperpigmentation, bumps/nodules and neuropathy were reported in the studies, and most events resolved over the course of months. Swelling, pain, edema, hair loss, altered sensation and palpable bumps were the most common and reported symptoms in the majority of the studies.

Discussion

Microwave-based devices have been gaining in popularity for the treatment of axillary hyperhidrosis because of its non-invasive, simple features. Although this innovative approach has demonstrated promising results in previous clinical studies, its efficacy still varies between different studies. In the current review, we systematically analysed the clinical evidence of the impact of microwave technology on the objective and subjective efficacy among patients with axillary hyperhidrosis. In the included studies, points of the HDSS and gravimetric assessment exhibited an obvious decrease, which displayed improvement in axillary hyperhidrosis in both the objective and subjective results. These results may be caused by a decrease in the number of sweat glands, which were revealed in the histology results of the studies mentioned above. However, one study showed no statistical differences in the gravimetric assessment after 30-day follow-up visits (8). This may result from a different system of a microwave device, ethnic groups and a lack of temperature control during the assessment. Nevertheless, most studies exhibited both a short-term and a long-term improvement in axillary hyperhidrosis. In addition to axillary hyperhidrosis, odour evaluation and patient satisfaction were also performed in some studies that we included. Patients with malodour obtained improvement after treatment, and most patients were satisfied with the procedure. Therefore, microwave-based device treatment achieved improvement in both axillary hyperhidrosis and malodour. More importantly, the patients were satisfied after the treatment.

Despite the good efficacy of microwave-based device treatment, there were some common adverse effects reported in the studies we analysed. Swelling, pain, edema, hair loss, altered sensation and palpable bumps were common and reported in the majority of the studies. There was also substantial variation in the duration of the mild effects and type of adverse effects between the various studies; however, most of the adverse events resolved over time. These effects may be due to a different system of microwave devices and protocols used, different ethnic groups and the presence of subjective bias. There were no procedure-related serious adverse events reported in the studies; only one person had treatment-related neuropathy, which resolved with six months (9). Overall, the microwave-based device is a safety choice with mild side effects for the treatment of axillary hyperhidrosis.

There are many treatments reported in the literature for axillary hyperhidrosis, such as temporary treatments (e.g., botulinum toxin and topical agents) and permanent treatments (e.g., excision with thoracic sympathectomy, subcutaneous shaving, tumescent liposuction with or without curettage and lasers used on the skin surface or subdermis) (4,13). Injections of botulinum toxin and topical agent treatments have also been found to be effective, but one should be treated repeatedly to maintain dryness. The effectiveness lasts between only four to nine months for botulinum toxin treatments (14–17), and topical agents should be applied within days based on the type of treatment used (18–20). Permanent treatment modalities exhibiting long-term efficacy have been explored in past studies, while the obvious adverse effects may lead these treatments to be unfavourable for many patients. Thoracic sympathectomy may lead to compensatory hyperhidrosis, and there was a study reporting that 88% of patients suffer from compensatory hyperhidrosis (21). Moreover, subcutaneous suction curettage is associated with pain, infection, hematoma and damage to the brachial plexus (22). Another surgical method can also result in infection, scars and restricted arm movement (23). Specifically, the Nd:YAG laser has been shown to reduce axillary hyperhidrosis recently; however, like other invasive treatments mentioned above, this procedure is associated with significant scarring, retraction and long recovery times (6,24). With long-term efficacy and mild side effects mentioned in this review, the microwave-based device may be a better choice for the treatment of axillary hyperhidrosis.

This systematic review has found that the microwave device is effective in the long-term follow-up for both subjective and objective results, with mild adverse events and high satisfaction. However, there are some limitations in the current review, and our study highlighted several areas of future research directions: (1) in the existing studies, there was only one RCT, and all the studies are limited by a small sample size. To further assess the value of microwave-based device treatment in axillary hyperhidrosis, more multicentre RCTs with a large number of subjects are needed, particularly compared to the existing treatments; (2) not all studies included in our review mentioned temperature, mood or other factors affecting sweating when evaluating the results of the studies. Thus, additional research should be conducted using the same conditions in the future; (3) the longest follow-up duration in the studies we investigated was only one year, while axillary hyperhidrosis is a long-term disease. Thus, clinical trials with longer follow-up periods are required in future studies.

Conclusion

Since Johnson and colleagues (7) introduced the use of a microwave device for the long-term treatment of axillary hyperhidrosis, microwave treatment has become increasingly important for the treatment of axillary hyperhidrosis. This systematic review indicated that microwave-based device therapy of axillary hyperhidrosis achieved long-term efficacy in both subjective and objective results. Moreover, mild side effects were presented, and most patients were satisfied with this treatment. However, the clinical research on this topic remains extremely limited. Further investigation is necessary to determine the long-term efficacy and safety of this treatment modality.

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Disclosure of interests

The authors have no conflict of interest or financial ties to disclose.

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